Jiwon Kim

+82 10-6361-9410 | kim.jiwon@ewha.ac.kr | jwkim-research.github.io

Education

Ewha Womans University, Seoul, South Korea — Sep 2025–Present
M.A. in Business Administration (Management Decision Science), expected Aug 2027
GPA: 4.23/4.30

Ewha Womans University, Seoul, South Korea — Aug 2025
B.B.A. in Business Administration; B.S. in Statistics (Double Major)

Bern University of Applied Sciences, Bern, Switzerland — Feb–Jul 2024
Exchange Student in Business

Grants

Master's Student Research Fellowship, National Research Foundation of Korea (NRF) — Sep 2026–Aug 2027
Principal Investigator | KRW 12,000,000
Two-Stage Matheuristic for Transportation Volume Reallocation

Publications and Manuscripts

Published Journal Article

Kim, J. and Min, D. (2026). "Assessing the Validity of Prompting Large Language Models for Vehicle Routing Problems." *Korean Management Science Review*, 43(2), 29–47. DOI

Working Paper

Moon, J., **Kim, J.**, Lah, Y., Jin, H., Han, Y., and Kang, Y. (2026). "Beyond Averages: Evaluating LLMs on Human Survey Replication at the Distributional Level." arXiv:2606.09013.

Kim, J., Lee, S., and Kang, Y. "Two-Stage Matheuristic Framework for Transportation Volume Reallocation."

Conference Presentations

Kim, J. (presenter), Lee, S., and Kang, Y. "Two-Stage Matheuristic Framework for Transportation Volume Reallocation."

- *INFORMS Annual Meeting*, San Francisco, CA, USA, November 4, 2026. (Scheduled)
- *Spring Joint Conference of KORMS, KIIE, and KSS*, Gyeongju, South Korea, June 5, 2026.

Patent

"Hierarchical Synthetic Persona Generation Method Using LLM and Multimodal Inputs." Korean Patent Application No. 10-2025-0212331. Co-inventor; pending.

Research Experience

AI Applications Lab (AAPL), Ewha Womans University — Sep 2024–Present
Advisor: Prof. Yuncheol Kang

Graduate Researcher — Sep 2025–Present
Undergraduate Research Intern — Sep 2024–Aug 2025

Transportation Volume Reallocation in Automotive Supply Chains
With Prof. Seokgi Lee, Youngstown State University

- Designed and implemented a two-stage matheuristic combining unused-capacity-based MILP initialization with MILP-based large neighborhood search under the full cost objective.
- Conducted computational experiments across problem sizes and item-volume settings, comparing solution quality against the full MILP solved with Gurobi under equal time budgets.
- Achieved lower total costs than the full MILP approach on all largest tested instances within a six-minute computational budget.

LLM-Based Human Response Simulation

- Contributed to experimental execution and the collaborative development of persona and prompt designs for LLM-based consumer simulation.
- Interpreted experimental results and synthesized findings across experimental conditions to organize key conclusions about the capabilities and limitations of LLM-based consumer simulation.

Research Assistant, School of Business, Ewha Womans University Sep 2025–Mar 2026
Advisor: Prof. Daiki Min

Prompt-Based Vehicle Routing with Large Language Models

- Developed and implemented a prompting methodology for generating vehicle routes using zero-shot, meta-prompting, iterative improvement, and few-shot approaches.
- Conducted computational experiments on CVRP and VRPTW benchmarks, evaluating solution feasibility, objective values, and structural similarity between routes.
- Analyzed limitations associated with problem size and constraint complexity and identified potential uses of LLMs as complementary routing decision-support tools.

Teaching Experience

Teaching Assistant, Ewha Womans University Fall 2025–Present

- **Operations Management**: Fall 2025, Spring 2026, Fall 2026
- **Statistics for Business**: Fall 2025, Spring 2026
- **AI and Optimization for Business**: Spring 2026
- **Management Science Theory**: Spring 2026

Responsibilities included grading assignments and exams and proctoring examinations.

Research Interests and Skills

Interests: Combinatorial optimization, sequential decision making, matheuristics, logistics and supply chain management, decision making under uncertainty.

Programming and tools: Python, R, Gurobi, OR-Tools, PyVRP, PyTorch.

Relevant coursework: Optimization Theory, Sequential Decision Making, Bayesian Statistics, Operations Management, Supply Chain Management, Data Science.